

```

1 public class N_Queens {
2
3     public static void main(String[] args) {
4         int n = 4;
5         int[] arr = new int[n];
6         N_Queens nQueens = new N_Queens();
7         nQueens.NQueens(arr, 0, n);
8     }
9     public boolean place(int k, int i, int[] arr){
10         for(int j = 0; j < k; j++){
11             if((arr[j] == i) || (arr[j]-i) == (j-k) || arr[j] - i == k - j){
12                 return false;
13             }
14         }
15         return true;
16     }
17     private void printSolution(int[] arr, int n) {
18         for (int i = 0; i < n; i++) {
19             for (int j = 0; j < n; j++) {
20                 if (arr[i] == j) {
21                     System.out.print("Q ");
22                 } else {
23                     System.out.print(". ");
24                 }
25             }
26             System.out.println();
27         }
28         System.out.println();
29     }
30     void NQueens(int[] arr, int k, int n){
31         for(int i = 0; i < n;i++){
32             if(place(k,i,arr)){
33                 arr[k] = i;
34                 if(k == n-1){
35                     printSolution(arr,n);
36                 }
37                 else{
38                     NQueens(arr, k+1,n);
39                 }
40             }
41         }
42     }
43 }
44
45
46
47 /*
48 Output:
49
50 . Q . .
51      ~

```

51 . . . Q
52 Q . . .
53 . . Q .
54
55 . . Q .
56 Q . . .
57 . . . Q
58 . Q . .
59
60 * */